

Supplementary Material

Thermal representation as a decision variable in heat-adaptive tourism opportunity screening: evidence from a Madrid pilot

All values below are exact, drawn from the locked project outputs; none is recalculated. Each item is included because it supports reproducibility or supplies exact detail behind a main-text figure, without distracting from the tourism-management argument.

Figure S1. Model-derived UTCI field at 12:00, 15:00 and 18:00

Model-derived UTCI across the day (SOLWEIG/UTCI, 2.5 m; shared scale)

White contours mark the 32 °C and 46 °C category breaks · white rings = outdoor assets · at 12:00 the field is ≥ 32 °C everywhere modelled

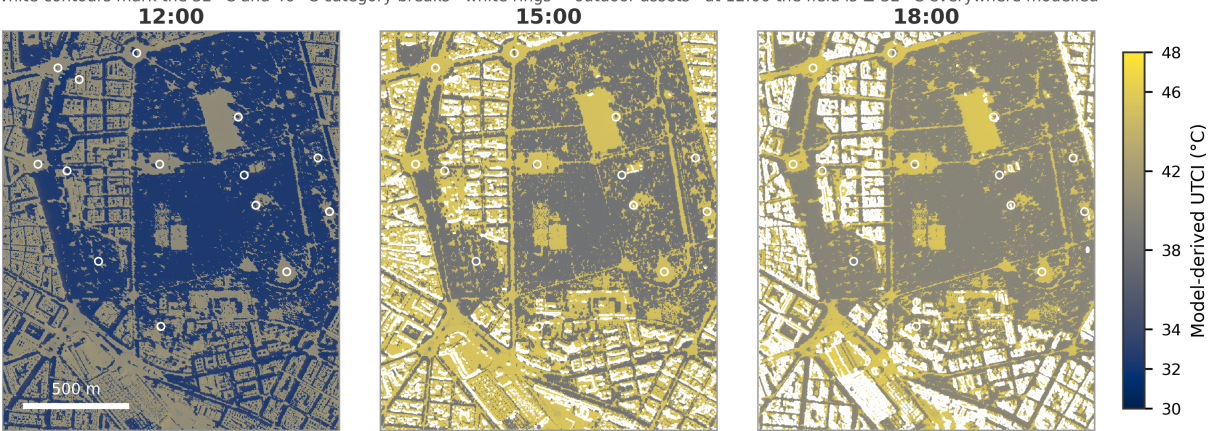


Figure 1: Figure S1 — Model-derived UTCI field at 12:00, 15:00 and 18:00.

Three-timestamp small-multiple of the **model-derived** UTCI field (SOLWEIG/UTCI, 2.5 m), on a single shared colour scale, with the 32 °C and 46 °C category breaks contoured and the 14 outdoor assets overlaid. Illustrates that the modelled radiant field is spatially and temporally structured, and that at 12:00 the field is ≥ 32 °C everywhere modelled (the mechanism behind the noon divergence in Figure 2). UTCI is model-derived, never observed. File: outputs/publication/figures/SFIG01_UTCI_FIELD_v0.1.{pdf,svg,png}. (Moved from the main text because the outdoor feasibility category is single-valued at the pre-registered 10 m buffer-mean, so the spatial texture is not the decision variable.)

Table S1. Thermal-method reclassification — exact rates

Source: outputs/tables/proxy_vs_physical_comparison.csv. Unit: outdoor asset $\times$ timestamp observation (n = 42).

Breakdown	Group	Value	n
Overall reclassification rate	all	33.3%	42
Direction	physical more restrictive than proxy	9	42
Direction	physical less restrictive than proxy	5	42
Direction	agreement (unchanged)	28	42
By timestamp	12:00	64.3%	14
By timestamp	15:00	0.0%	14
By timestamp	18:00	35.7%	14
By morphology	attraction exterior	33.3%	9
By morphology	park or garden	33.3%	18
By morphology	plaza or hardscape	33.3%	12
By morphology	street corridor	33.3%	3

“More/less restrictive” denotes the direction of classification divergence between the two methods, not the correctness of either; neither method is field-validated.

Table S2. Decision confidence and tested-uncertainty detail

Sources: outputs/tables/solar_forcing_sensitivity.csv, data/processed/phase2_2_decision_confidence.csv
Envelope = [min, max] UTCI (°C) over the tested realizations.

(a) Solar-forcing sensitivity (decisions changed vs clear-sky baseline, of 42):

Realization	Decisions changed	%
Satellite-derived irradiance (EUMETSAT CM SAF)	1	2.4%
–10% GHI	0	0.0%
–20% GHI	0	0.0%

(b) Confidence distribution (of 42 outdoor observations): ROBUST 35 (83.3%), BOUNDARY 6 (14.3%), UNSTABLE 1 (2.4%).

(c) The six BOUNDARY and one UNSTABLE observations (near the 46 °C boundary):

Asset	Timestamp	Class	Envelope (°C)
A15	15:00	BOUNDARY	42.4–44.5
A16	15:00	BOUNDARY	43.7–45.1
A24	15:00	BOUNDARY	44.1–45.1
A25	15:00	BOUNDARY	44.0–45.0
A15	18:00	BOUNDARY	42.8–45.2
A16	18:00	BOUNDARY	43.9–45.9
A24	18:00	UNSTABLE	44.2–46.0

Tested uncertainty covers solar forcing (four realizations, all rows) plus targeted canopy geometry (two assets); it is a lower bound on total uncertainty and is not a validation of the modelled field.

Table S3. Exclusion-reason frequency across the eight scenarios

Source: `outputs/tables/phase3_exclusion_reasons.csv`. Every excluded candidate carries exactly one machine-readable reason (defined in Table 2).

Exclusion reason	Count across scenarios
ACCESSIBILITY_CONSTRAINT	93
CLOSED_AT_TIMESTAMP	43
OUTDOOR_EXPOSURE_TOO_HIGH	20
NO_MEANINGFUL_THERMAL_IMPROVEMENT	2
INSUFFICIENT_EVIDENCE	1

The 23 options reported in the main text as removed “on thermal/evidence grounds” are the `OUTDOOR_EXPOSURE_TOO_HIGH` + `NO_MEANINGFUL_THERMAL_IMPROVEMENT` + `INSUFFICIENT_EVIDENCE` removals attributable to each scenario’s source-relative screen (Table 3); accessibility and opening-hours exclusions are counted separately above.

Table S4. Opening-hours provenance (summary)

Source: `data/processed/phase3_osm_opening_hours_raw.csv`, `src/phase3_build_catalog.py` (`DOCUMENTED_HOURS`). Of 27 assets, opening hours were harvested from OpenStreetMap tags for 11 and filled from documented institutional schedules for 16; each carries a source string and an evidence-completeness flag. Hours were documented in 2026 and applied to the 2023 study day (temporal-alignment limitation, Section 5.5). Real Monday-in-August closures used include Museo Naval, Antropología and Artes Decorativas, Real Fábrica de Tapices, Thyssen (16:00 close), and the Real Observatorio (guided visits Fri-Sun only). A per-asset table with source strings is available in the repository.

S5. Software and reproducibility

Physical modelling used the official standalone `solweig` package (version 0.1.0b92) under a dedicated Python 3.12 environment; the geospatial and screening pipeline ran under the project’s main Python. UTCI was computed by the package’s own built-in module. All reported tables and figures regenerate from the locked inputs via the project scripts (`src/*.py`; figure scripts in `outputs/publication/figures/`), with per-figure numeric assertions and SHA-256 hashes recorded in `docs/PHASE5_3B_RENDER_MANIFEST.csv`.

Model parameters were left at documented package defaults (Section 2.4); no parameter was tuned to a target result. [PUBLIC REPOSITORY URL / RELEASE DOI TO VERIFY.]